



CSE-476: Introduction to Natural Language Processing  
Quiz 3

## Multiple Choice Questions (Single correct)

*Instructions: Select the correct option with a short explanation.*

- Q1. In a standard RNN, what is the primary purpose of the “hidden state”?
- (A) To store the final output of the entire sequence.
  - (B) To act as a memory that carries information from previous time steps
  - (C) To calculate the loss function for the current word
  - (D) To normalize the input embeddings
- Q2. Why is the Softmax function typically used in the final layer of a Neural Language Model?
- (A) To ensure the output values sum to 1, representing a probability distribution
  - (B) To prevent the gradients from exploding
  - (C) To increase the speed of the training process
  - (D) To allow the model to handle variable-length sequences
- Q3. What is a key limitation of feed-forward neural networks for NLP tasks?
- (A) They cannot use embeddings
  - (B) They assume fixed input size and struggle with temporal order
  - (C) They cannot be trained with gradient descent
  - (D) They do not use activation functions
- Q4. What is the purpose of the “Forward Pass” in training?
- (A) To update weights based on the loss
  - (B) To calculate the predicted output and compute the loss
  - (C) To unroll the network into a sequence
  - (D) To calculate the gradient for each parameter
- Q5. What is true about the weights  $U$ ,  $V$ , and  $W$  across different timestamps in an RNN?
- (A) They are randomly re-initialized at every step
  - (B) They are reused across timestamps

- (C) They increase in value as the sequence progresses
  - (D) Only the output weight  $V$  is reused
- Q6. Suppose all activation functions in a 3-layer neural network are linear. What is the effective model equivalent to?
- (A) A deeper network with more representational power
  - (B) A single linear model
  - (C) A recurrent neural network
  - (D) A decision tree
- Q7. When training an RNN for language modeling using teacher forcing, the model predicts:
- (A) The next word given the predicted history
  - (B) The next word given the ground-truth history
  - (C) The current word given itself
  - (D) All words simultaneously
- Q8. In an RNN, the hidden state at time  $t$  depends on:
- (A) Only the input at time  $t$
  - (B) Only the hidden state at time  $t - 1$
  - (C) Both the input at time  $t$  and the hidden state at time  $t - 1$
  - (D) All future inputs
- Q9. Consider a neural network layer defined as:  $h = f(Wx)$ . The input vector  $x$  has shape  $(n_0 \times 1)$ , and the output vector  $h$  has shape  $(n_1 \times 1)$ . What must be the shape of  $W$ ?
- (A)  $(n_0 \times n_1)$
  - (B)  $(n_1 \times n_0)$
  - (C)  $(n_0 \times 1)$
  - (D)  $(1 \times n_1)$
- Q10. What is the main role of backpropagation in neural network training?
- (A) To compute predictions during inference
  - (B) To calculate gradients of the loss with respect to parameters
  - (C) To initialize weights randomly
  - (D) To normalize input data

*End of Paper*